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RESEARCH

When to Re-Plan: Subgoal Persistence in Hierarchical Latent Reasoning

ACCEPTED · **Sole Author** · 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents @ ICML 2026 · *Seoul, South Korea*

Workshop organized by researchers from IBM Research, OpenAI, and Harvard.

Jan 2025 – May 2026 · *Paper* · *Code / SCOUT*

- Studied how hierarchical latent reasoners should maintain and revise **medium-horizon intent** while computation unfolds recurrently in hidden state.
- Extended the Hierarchical Reasoning Model with a **manager-worker architecture** in which a high-level module produces a latent subgoal that persists across multiple reasoning steps.
- Found that revising the subgoal at every step can disrupt coherent reasoning, while moderate persistence allows the model to build on a stable direction. The work frames persistent subgoals as **reusable compositional structure for long-horizon reasoning and planning**.

A Score Is Not a Policy: Measuring Whether a Supervisory Module's Decisions Are Worth Making

Sole Author · Preprint · *Under Review*

May 2026 – Present · *Paper*

- Extended the persistence question from fixed schedules to **learned meta-level control**: can a supervisor use a reasoner's internal state to decide when an active commitment should persist or be revised?
- Developed an evaluation framework that distinguishes whether a supervisory score depends on model state, whether it changes system behaviour, and whether those decisions improve outcomes beyond strong fixed strategies.
- Found that an apparently adaptive supervisor can largely follow timing structure without learning a useful decision policy. The result motivates stronger evaluation of supervisory modules in compound agentic systems, where producing a plausible score is insufficient unless acting on it creates measurable value.

SAINT

Independent Research Venture 2026 – Present

- Exploring how AI agents can decide whether additional reasoning is worth the computation it requires: more time, another attempt, or a change in strategy does not necessarily produce a better result.
- Building on research on subgoal persistence and adaptive revision, investigating methods for allocating reasoning effort according to its expected value, with planned experiments in coding agents working across real software repositories.
- Selected for **Accel AI Innovate: Paris**, Accel's Future Founders hackathon for AI builders, hosted with Iterate (October 2026).
- Participating in the **SpaceXAI Macau Community Hackathon** at Macau Tower, Macao, in Track 2: **Physical AI** × **Grok Bot** (November 2026).

RESEARCH SERVICE

Invited Reviewer · NeurIPS 2026 Workshop on Responsible Use of Meta-Agents

Nominated by the Area Chair to review work on agents that design, optimize, supervise, train, or evaluate other agents, including agent harnesses and meta-level evaluation.

Advised by Christopher D. Manning, with organizers from Stanford, Carnegie Mellon, Google DeepMind, and Northeastern.

Reviewer · 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents @ ICML 2026

Reviewed research on compositional learning, reasoning, interpretability, and agentic systems.

PROFESSIONAL EXPERIENCE

Propero Consulting
Pune, Maharashtra, India

Software Developer · Machine Learning & Applied AI Aug 2022 – Dec 2024

- **Led the design and engineering of ShopiBot**, a domain-specific AI system supporting Shopify developers with technical questions, client requirements, and implementation guidance; translated an early product idea into retrieval, orchestration, and evaluation workflows.
- Studied and implemented retrieval approaches including dense semantic search, **RAPTOR-inspired hierarchical retrieval**, and corrective and adaptive RAG; combined query analysis with routing across retrieval pipelines to improve access to Shopify technical context.
- Built agentic workflows with knowledge graphs, LangChain planners, multi-tool reasoning, self-reflection, and LLM tool calling; explored **Sema4AI / Robocorp** for developer automation and created technical demos for prospective startup partnerships.
- Improved production reliability through automated evaluation, streaming inference, vector-database sharding, and caching, reducing end-to-end latency from **53s to 20s**. Earlier work included CLIP-guided generation and automated checkout-flow testing using DOM analysis and feedback-driven methods.

Unify · YC-backed
London, UK / Remote

LLM Engineer Contributor Jul 2024 – Sep 2024

- Selected for Unify's contributor programme, collaborating with the core team on a community-built **unified LLM query router** across model providers.

EDUCATION

Northeastern University
Boston, MA

M.S. Artificial Intelligence · Machine Learning concentration Admitted Fall 2026

Khoury College of Computer Sciences · Spring 2027 start option

International Impact Award · Inaugural Cohort Scholarship · 30% tuition scholarship

College of Technology, GBPUA&T
Pantnagar, India

B.Tech, Electrical Engineering · 73.2% · First Division 2017 – 2021

TECHNICAL SKILLS

Reasoning & ML: Hierarchical and recurrent latent reasoning, *adaptive computation*, supervisory / meta-level control, test-time computation, hierarchical and feudal RL, PyTorch, Hugging Face Transformers

LLM & Agentic Systems: Dense retrieval, RAPTOR, corrective/adaptive RAG, LangChain, LlamaIndex, tool calling, LangSmith, CLIP, diffusion models

Engineering & Infrastructure: Python, C, C++, CUDA, Docker, AWS (S3, Lambda, EKS), Pinecone, Chroma, mixed precision, gradient checkpointing, single-node multi-GPU training